

Büchi Automata and ω -regular Languages

A **Büchi Automaton** is a type of NFA with no ϵ -transitions that accepts inputs in Σ^ω , i.e. infinite length inputs.

Σ^ω represents any infinite word over alphabet Σ

Σ^* represents any finite word over alphabet Σ

A Büchi Automaton accepts an input string $w_1, w_2, \dots \in \Sigma^\omega$ if there exists a path through the machine (starting at a starting state and following the transition function at each step) such that there is an infinite number of accepting states in this path. Since FSMs have a finite number of states, this means that at least one state in the BA must be visited infinitely often.

A language is **ω -regular** iff it is recognised by some BA.

Operations:

Union: $A \cup B$, A and B ω -regular

Concatenation: AB , A is regular, B is ω -regular

Two ω -regular languages cannot be concatenated as they deal with infinite length words.

Similarly, A cannot be ω -regular with B regular, since we never get to B .

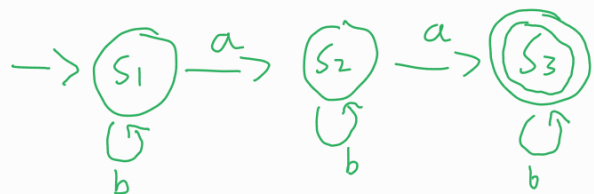
ω : A^ω - an infinite sequence of words from A , where A is regular and does not contain the empty word.

All these operations produce an ω -regular language.

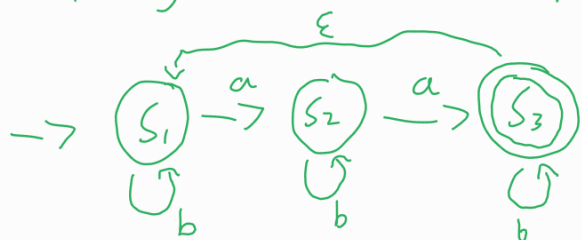
Example

L is a regular language recognised by the following NFA:

$$L = b^* a b^* a b^*$$



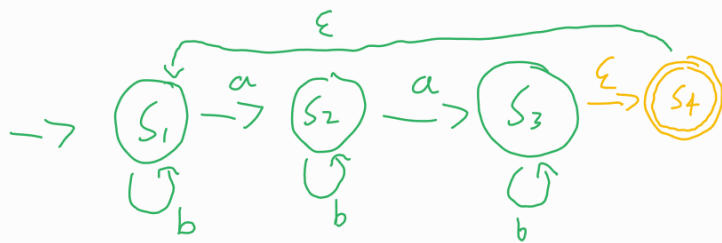
The following is not a valid Büchi for L^ω :



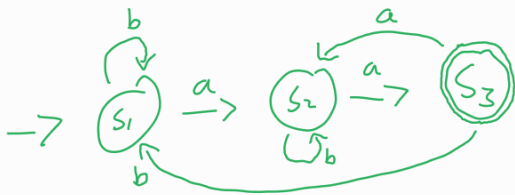
① Büchi automata cannot have ϵ -transitions.

② This automata accepts the string $a a b^\omega$ which is not in L^ω as it is not an infinite sequence of words in L , but visits accepting state S_3 infinitely often.

Fix for (2) - The BA is forced to go back to the start after reaching the accept state.



Fix for (1) and (2):



Note: the initial invalid Buchi accepts $\text{Lim } L$.

$\text{Lim } L$ is the language $\{a \in \Sigma^\omega \mid a \text{ has infinitely many prefixes in } L\}$

↓
this means that as you read the infinite sequence, there are infinitely many points such that the string read so far is a valid string in L .

$L^\omega \subseteq \text{Lim } L$, but $\text{Lim } L$ also includes aab^ω etc $\rightarrow$ doesn't require infinitely many words from L .

An ω -regular language is the limit of some regular language iff some deterministic BA recognises it.